MDP-based Reliability Analysis of an Ambient Assisted Living System

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Abstract. The proliferation of ageing population creates heavy burdens to all industrialised societies. Smart systems equipped with ambient intelligence technologies, also known as Ambient Assisted Living (AAL) Systems are in great needs to improve the elders' independent living and alleviate the pressure on caregivers/family members. In practice, these systems are expected to meet a certain reliability requirement in order to guarantee the usefulness. However, this is challenging due to the facts that AAL systems come with complex behaviours, dynamic environments and unreliable communications. In this work, we report our experience on analysing reliability of a smart healthcare system named AMUPADH for elderly people with dementia, which is deployed in a Singapore-based nursing home. Using Markov Decision Process (MDP) as the reliability model, we perform reliability analysis in three aspects. Firstly, we judge the AAL system design by calculating the overall system reliability based on the reliability value of each component. Secondly, to achieve the required system reliability, we perform the reliability distribution to calculate the reliability requirement for each component. Lastly, sensitivity analysis is applied to find which component affects the system reliability most significantly. Our evaluation shows that the overall reliability of reminders to be sent correctly in AMUPADH system is below 40%, and improving the reliability of Wi-Fi network would be more effective to improve the overall reliability than other components.

Keywords: Reliability Analysis·MDP·Ambient Assisted Living·Sensitivity

1 Introduction

The rapid increase of ageing population in all industrialised societies has raised serious problems, e.g., creating enormous costs for the intensive care of elderly people. The Ambient Assisted Living (AAL) system, as a promising solution is designed to assist their independent living [16, 20]. In such systems, sensors and inference engines are widely used to perceive environment changes and user intentions. Applications and actuators are triggered accordingly to provide necessary assistance to the user. For example, if the system detects abnormal behaviours such as the user is showering for too long, a reminder will be prompted to advise the user finish showering. However, lack of reliability guarantees prevents AAL systems to be widely used. A failure of prompting a reminder could harm the user's life, e.g., a call-for-help alert failed to prompt when

the elder falls may leave him/her unattended for a long time leading him/her to death. Thus, it is essential to conduct reliability analysis and provide quantitative guarantees on the system before deployment.

An AAL system is considered reliable if the reminder service is correctly delivered to the right person at the right scenario. It is a challenging task to analyse the reliability of such systems. First of all, AAL systems are inherently complex. They are usually composed of multiple layers of software and hardware components which have limited capability and accuracy. Previous research [17, 18] reported that the often inherent inaccurate and unreliable low-level sensors are used to detect context information from the environment. This is probably due to cost efficiency considerations, i.e., low-capability but cheap sensors are selected due to budget constraints. Furthermore, AAL systems rely heavily on different types of wireless networks with different reliability. For instance, sensors transmits signals via Zigbee, a low-cost low-power wireless mesh network, while software components and actuators exchange messages using more reliable networks such as WLAN. Moreover, human errors, e.g., a user may forget to wear the RFID tags, could also cause the failure of the system. Since failures of such systems are unavoidable, it becomes critical to manage the reliability in an acceptable level.

Quantitative analysis of probabilistic systems gains great importance, especially for complex systems with non-deterministic behaviours. AAL systems are typically user centred so that their system behaviours are non-deterministic due to unpredictable user activities. Thus, we propose Markov Decision Process (MDP) as the modelling formalism for its support of both non-deterministic as well as probabilistic choices. Three general but highly important reliability issues are investigated. First, "what is the overall system reliability if reliability of all its components is known, considering all possible user behaviours, and unreliable factors?". This is the problem of *reliability prediction*. This question is to be answered necessarily before system deployment since end users would prefer to know how reliable the system is. Secondly, "what is the reliability required on certain components if there is a requirement on overall system reliability?". This is the problem of *reliability distribution*. Addressing this issue is useful because we can have specific quantitative requirements on selecting software and hardware components, whose quality are often cost sensitive. Last but not the least, to find the most critical components affecting the system reliability via *sensitivity analysis* is essential to improve the overall system reliability effectively with limited resource [5, 7]. For example, if a system is shown to be not reliable enough based on current component's reliability, it is desirable to prioritise the components such that reliability improvement of a higher priority component would result in more improvement on overall system reliability.

In this work, we applied the reliability analysis on a smart healthcare system, AMUPADH [4]. Our evaluation shows that its overall reliability is below 40%. In order for the system to reach a reliability of 0.4, each Wi-Fi network related node should have a reliability of at least 0.9 proved in the reliability distribution analysis. However, our analysis also concludes that it is impossible for the system reliability to reach 0.5 based on the current design. There is such a scenario that the system always fail to recognise who is the person doing the abnormal activity. Thus, half of the chances the reminder will be sent to a wrong person. Lastly, the sensitivity analysis suggests that improving

the reliability of Wi-Fi network would be more effective to improve the overall reliability. In the end, the analysis results are reported to AMUPADH designers who confirmed their consistency with the real data collected from the hospital. As a result, they redesigned the activity recognition rules and added more nodes in the Wi-Fi network to increase its reliability. Our experience in this case study demonstrates the usefulness of reliability analysis using probabilistic models. It is able to provide good estimation of system reliability and identify the critical component inside the system.

Related Works In the literature, there has been some work on analysing reliability of complex systems. Reliability analysis by modelling system architecture as Discrete-time Markov chain (DTMC) is first proposed by Cheung [5] in 1980. It has been applied in various case studies, e.g., Gokhale et al. [8] analysed a stochastic modelling tool, SHARPE by constructing a DTMC and found out the relation between system reliability and fault density per subsystem. Goseva et al. [9] performed reliability predicting and sensitivity analysis on a system of the European Space Agency. Wang et al. [19] analysed a stock market system by constructing DTMC and predicted the system reliability. However, to the best of our knowledge, there is no reliability analysis has been conducted on any AAL system which involves not only system reactions but also non-deterministic human behaviours. In such a complex system, probability distribution of transitions among system components are hard to obtain. Thus, we choose MDP over DTMC. Furthermore, most of the works are focusing on predicting reliability of exiting systems while we contribute more on finding the best solutions to improve system reliability via reliability distribution and sensitivity analysis.

The remainder of the paper is organised as follows. Section 2 introduces our case study system, AMUPADH, a typical AAL system designed for elderly dementia people's healthcare. Sections 3 demonstrates the process of modelling system in MDP using a typical scenario in AMUPADH. Techniques for reliability analysis are briefly introduced in Section 4 as well as the experiments and discussions. Section 5 concludes the paper and discusses future directions.

2 AMUPADH: An Ambient Assisted Living System for Elderly Dementia People's Healthcare

Dementia is a progressive, disabling, and chronic disease common in elderly people. Elders with dementia often have declining short-term memory and have difficulties in remembering necessary activities of daily living. However, they are able to live independently in assisted living facilities with little supervision. AMUPADH system is designed for this purpose by providing necessary assistance in the form of reminders.

AMUPADH is a project initiated in Singapore to design smart healthcare systems for monitoring and assisting the daily living of elderly people with dementia. This project started with three months' visits to PeaceHeaven[3] nursing home for collecting requirements. By observing the patients' daily life and interviewing nurses/doctors, two critical issues associated with dementia patients are raised which are sleeping behaviours in bedroom and showering behaviours in bathroom. With 21 months research

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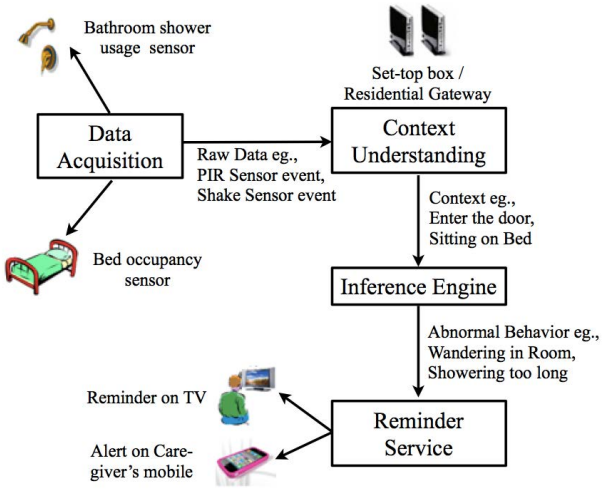


Fig. 1: An Overview of AMUPADH system design

and development focusing on providing assistance on these two scenarios, the system was finally deployed in the nursing home to be tested with real dementia patient users for 6 months. Our idea is inspired by this trial deployment.

Preliminary data collected in the trial shows that the importance of system reliability is underestimated. Unreliability caused by sensor/device failures, network issues and unforeseen human errors draws considerable attentions of the stakeholders.

2.1 System Components

The design of the system is shown in Figure 1. It mainly consists of three sub-systems, *Data Acquisition* component containing various sensors, the *Context Processing* and *Inference Engine* components based on first order logic rules and *Reminder System* for rendering suitable reminder services to the patients.

– **Data Acquisition** In the system, multiple sensors are deployed to acquire information from the home environment. For example, if someone turns on the shower tap, the shake sensor on shower pipe will be triggered and change its status to $Unstationary$. A signal is generated and then sent to the central system via a Zigbee network. AMUPADH adopts a multi-modal sensor[4] design for user monitoring. This is due to users' privacy concerns, video cameras are refused in bedrooms.
The PeaceHeaven nursing home has 13 separate Resident Living Areas (RLAs), each designed as an individual home-like environment. Selected rooms for AMUPADH system deployment are equipped with two/three beds with a shower facility.

[4] Multi-modal sensor also known as sensor fusion is the combining of sensory data from disparate sources such that the resulting information is more accurate than using the sources individually.

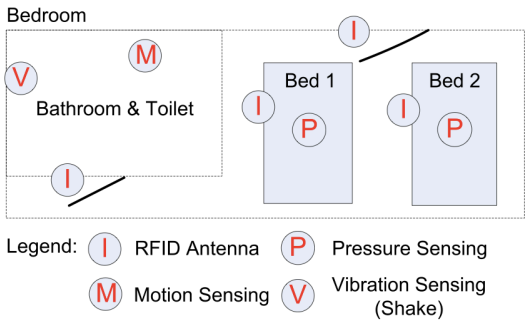


Fig. 2: Sensor Deployment in PeaceHeaven

Three rooms are selected for deployment; and each room is shared by 2 or 3 people. Figure 2 shows an exemplar sensor deployment for a twin shared room. The pressure sensor under a bed mattress is used for detecting sitting/lying behaviour, while the RFID readers are for detecting the identity of the person near the location. In the bath room, a motion sensor detects human presence in the room, while the vibration sensors are attached to water pipes and the soap dispenser for detecting their usages.

– **Context Processing and Reasoning** Upon receiving a sensor signal, the central system translates it into low-level context $sensor\ events$ i.e., a signal $unstationery$ from shake sensor on shower pipe is translated to "Shower Tap On". Different low-level contexts are provided from different sensors. They are aggregated in the inference engine for reasoning and generating high-level contexts, $activities$. This task is performed by evaluating predefined activity recognition rules based on prior knowledge of user behaviours. A typical rule is like: if shower tap is on and lasts for 30 minutes, at the meantime a PIR sensor detects movements of someone in the washroom; an abnormal behaviour, $showering\ for\ too\ long$ is recognised, then a message will be sent to the server indicating some patient is in the abnormal state of showering for too long.
  The messages are sent out via a shared bus within the central system. Note that, AMUPADH aims for a multi-user sharing environment which is a challenging topic in the activity recognition area. In fact, it is not only important to know which activity is being carried on but also who is doing this activity. This adds complexity to define the activity recognition rules. In the case that if the patient's identity is missing in a rule, the activity could be recognised for a different person, causing a subsequent reminder to be prompted to a wrong patient. Our previous work [15, 14] discussed this issue in details. In this work, when defining the reliability of the inference engine, we take this factor into account.
– **Reminding System** The reminding system listens to the messages sent from the inference engine and decides which reminder service to render. For example, upon receiving the message $Activity.error.ShowerTooLong.patientA$, the system will invoke the service of playing a preloaded sound reminder on bluetooth speaker lo-

cated in the shower room correspondingly. In this case, the message is transferred via bluetooth technology. In general, different message transmitting technologies are used for different rendering devices. For instance, for reminders on mobile phones, messages are transmitted through 3G network, while for iPad case, a small home wide Wi-Fi network is used.

2.2 Six Reminding Scenarios

In AMUPADH system, there are six reminding scenarios targeting at providing assistance for six abnormal behaviours of elderly patients with dementia.

- **Using Wrong Bed (UWB)** Since a room in the PeaceHeaven nursing home is shared by 2-3 patients, some of them, especially the new residences, tend to lie on a bed without recognising whether it is his/her own bed. This behaviour is detected by the bed pressure sensor and RFID reader. The reminder will be prompted by a Bluetooth speaker beside bed or an iPad on the wall asking the patient to leave the bed.
- **Sitting on Bed for Too Long (SBTL)** Some of the agitated patients often have sleeping problems. They are easily bothered and irritated by the environment. A typical symptom is that the patient will get up at midnight and sit on the bed for very long time until assisted by nurses/caregivers. The abnormal scenario is captured by bed pressure sensor and a timer in the mini-server. A reminder will be prompted using similar devices as UWB scenario whispering the person to sleep or send an alert to nurse's PC console/ mobile phone.
- **Shower No Soap (SNS)** Due to memory loss, dementia patients constantly forget the normal steps of performing an daily activity. In the taking shower activity, the patient could forget what to do next right after the shower tap is turned on. It is reported by the nurses that some of the patients finish the shower very fast without applying soap. Concerned about the personal hygiene, patients presenting this behaviour need help. Vibration sensors on the shower pipe and soap dispenser are used to capture the activity. A reminder instructing the person to use soap will be prompted by a Bluetooth speaker.
- **Showering for Too Long (STL)** Similar to the SNS scenario, some patients will stand under the shower head for a long time. This is a critical issue that exposing in the water for a long time could cause the patient faint. If not helped immediately, it can even cause death. Similar sensors and devices are used as SNS scenario to help the patient stop showering.
- **Tap Not Off (TNO)** It is often the case that dementia patients forget to turn off the tap after showering. In order to save water and energy, this scenario is detected by a RFID reader, a motion sensor and two vibration sensors. A reminder is prompted on proper device according to patient location asking someone to turn off the tap.
- **Wandering in Washroom (WiW)** Caused by memory loss, it is possible for the patient to forget at any step during showering. Thus, a wandering behaviour is also typical and patients need assistance in such cases. The wandering behaviour is monitored by a RFID reader and a motion sensor in the washroom. A leave-washroom reminder will be prompted to the person.

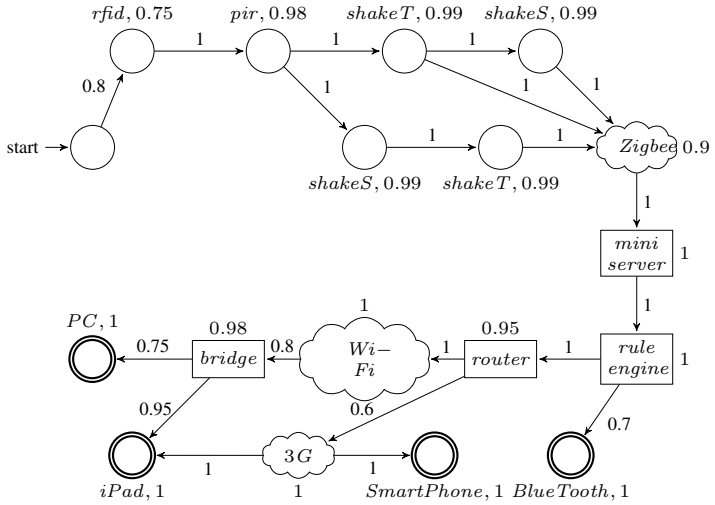


Fig. 3: Bathroom Scenario- TNO: Tap Not Off

In fact, taking shower turns out to be the most concerned issue of nursing elderly patients with dementia. In PeaceHeaven, the nurses need to monitor the showering activity of every patient. Considering the ratio of nurses to patients is 1:15, it creates a heavy burden to nurses. To alleviate the problem, a two-level reminding solution is provided in AMUPADH. When the system recognises an abnormal behaviour, it will prompt a reminder to the patient. If the problem remains, an alert will be sent to the nurse's mobile phone or PC console to raise her attention.

3 Modeling AMUPADH System

AAL systems are user driven such that the system behaviour contains non-determinism due to the unpredictable user behaviours. Thus, MDP is chosen as the modelling formalism in this work. Compared to DTMC, MDP allows us to capture both probabilistic and nondeterministic behaviours. A central issue in the AMUPADH system modelling is that when to use nondeterministic choices over probabilistic choices. In general, *probabilistic choices can be viewed as informed nondeterministic choices*. That is, we use a nondeterministic choice when we have no definitive information on how the choice is resolved. For instance, if all we know is that there are two different outgoing transitions after executing a component C, we model the two transitions using a nondeterministic choice. If the choice is made locally, after testing C systematically, we learn the frequency of each outgoing transition and we can model C with a probabilistic choice.

In practice, it turns out to be unrealistic to model all the scenarios using one MDP model considering the complexity and readability. Thus, we split the model into six models according to different scenarios by duplicating the same components. In the

following, we shall explain the modelling processes. Scenario TNO is taken as an example for its richness of involved components. Generally, as shown in Figure 3, there are three major elements in an MDP model for AAL systems, i.e., the nodes, the transitions and the reliability values.

Nodes Typically, in an AAL system, the sources of unreliability could be failure of sensors and network devices, error in softwares and connection loss/transmission failure in networks. Thus, in an MDP model of an AAL System, nodes are the abstraction of sensors, software components and network devices. To decide which device/component is necessary to be modelled, we need to analyse the activity recognition rules. In TNO case, four sensors are used for recognising this behaviour as introduced in Section 2.2. Besides, there are multiple choices of playing this reminder, e.g., playing on an iPad where reminder command is received via Wi-Fi network or on a Smart Phone through 3G network. Thus, the four sensors, iPad, smart phone, Wi-Fi network and 3G network need to be included in the model as nodes. Similarly, the Zigbee network, mini server and the rule engine need to be modelled as well.

In Figure 3, circle nodes denote sensors, square nodes denote hardware devices and cloud shape nodes denote networks. Double circled nodes are accepting nodes representing a reminder is successfully delivered. The different shapes of nodes are used to show the different types of components. In the MDP model, they are treated the same.

Transitions In AAL systems, there are usually two types of relations between nodes, happen-before and message-forwarding relations. Happen-before relation usually exists among sensors saying that some sensor is triggered earlier than the others. It is able to be derived from analysing the temporal relations between sensors according to their spatial distribution. For example, in Figure 2, the RFID reader near the bedroom door is triggered earlier than other sensors assuming the system starts with all users outside. Thus, in the MDP model, it should be placed in front of the rest of sensor nodes.

However, sometimes, the happen-before relation is not deterministic. For instance, for the model shown in Figure 3, there is no specific triggering orders between shake sensors on the tap and soap dispenser. Thus, we need to enumerate all the possible orderings. Besides, there is one rule deciding this abnormal behaviour based on shower-pipe vibration sensor only. Thus, there is a transition link from $ShakeT$ to $Zigbee$ making the ordering asymmetric. Our experience suggests that it is better to enumerate all the possible transition orders in the initial model, especially when there are multiple rules defined for recognising the same scenario.

As for message-forwarding relations, they are extracted from the system design. For example, in the TNO model, the messages are sent to the mini server via Zigbee network. Thus, a Zigbee node is placed between the sensors and mini server. The transitions between nodes denote the direction of message transmission. Similar methods are applied for the rest of the transitions.

Reliability and Transition Probability Labelling The final step is to label the nodes and transitions. Nodes are labelled with reliability values of the corresponding devices.

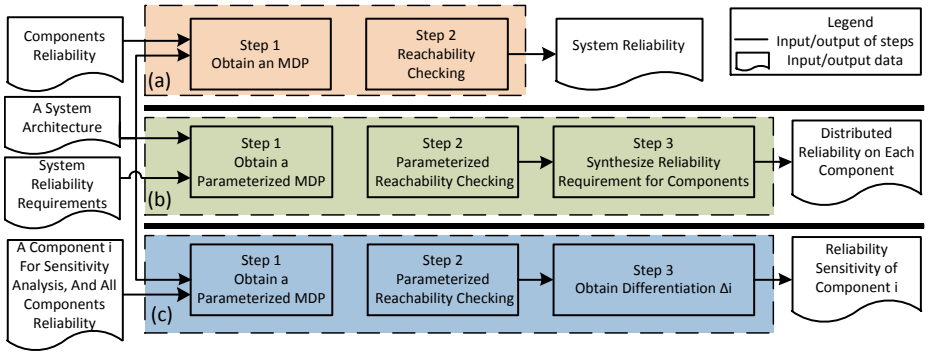


Fig. 4: Workflow: (a) Reliability Prediction; (b) Reliability Distribution; (c) Sensitivity Analysis

For transitions, there are different cases. At the initial node, the outgoing transitions usually representing the user behaviours. In the TNO case shown in Figure 3, there is 20% of time, the user will throw the RFID tags away (result drawn from an experiment conducted by the engineers). Thus, initially, there are only 0.8 probability leading to the next node. Additionally, the happen-before relations are usually non-deterministic choices with no specific probabilities due to randomness of user activities, thus by default, we assign the value 1. As for forwarding relations, due to the signal strength, transitions to/from network nodes have different reliability values. Transitions from Wi-Fi node to bridge node has the reliability of 0.8 since the bridge is placed on the wall outside the bedroom. The nurse PC in common area is further away from the bedroom, thus the transition from bridge to PC is as low as 0.75.

In our case study, these reliability values are provided by system engineers. Fortunately, AMUPADH system has been deployed in a real user environment for data collection. During the 6 months trial deployment and 3 consecutive months, 24 hours data collecting, the engineers are able to log every details of how the system works. By comparing to the ground truth (manually logged by nurses in the nursing home) and conducting statistical analysis, they are able to provide a good estimation of each component's reliability.

4 Reliability Analysis on AMUPADH: Experiments and Evaluations

4.1 Reliability Analysis Tool Framework

We make use our home-grown tool RaPiD [11] for reliability analysis which is specially designed for solving reliability problems. The reliability prediction problem can be solved conveniently using probabilistic model checking, whereas reliability distribution and sensitivity analysis is supported by extending RaPiD. We use the standard algorithms like value iteration or linear program solving [3] for calculating reliability. Many other probabilistic tools like PRISM [13] can be easily extended to perform the

verification. In this section, we briefly introduce the workflows of the reliability analysis including reliability prediction, distribution and sensitivity analysis, shown in Figure 4.

Reliability Prediction As shown in Figure 4(a), the reliability value of each component and an MDP model of a system are required for calculating the overall system reliability. Reliability prediction is equivalent to check the probability of the system never fails. It calculates the probability of reaching accepting nodes, $Pr(M, s)$ from an initial state to a goal state s on an MDP model M. A reliability range i.e., max. and min. reachability is produced since multiple reachable paths are created due to nondeterminism. Different methods have been developed to perform this task. Interesting readers are referred to [11] for more details. We remark that, unlike DTMCs approach, the result here is a probability range. The upper bound is the system reliability corresponding to the best scenario in AAL systems, whereas, the lower bound is corresponding to the worst scenario.

Reliability Distribution In addition to Figure 4(a), reliability distribution analysis shown in Figure 4(b) needs two more inputs: (1) a reliability requirement R on the overall system; (2) a parameterised MDP model M with weights $w_i x$ (denotes the reliability of component x has a weight w_i). Given a scheduler[5] δ, we can obtain the system reliability (i.e., $Pr(M, s)$) as a polynomial function of x only. Then the constraints on individual components are solved using numerical methods. RaPiD uses Newton's method, due to its fast convergence rate to the solution/root. It will calculate the lower bounds on x for finitely many schedulers [3] among which the maximum value gives us the minimum requirement on component reliability.

Sensitivity Analysis Sensitivity analysis requires all component reliabilities known in advance and an indication on which one of those components needs to be tested as shown in Figure 4 (c). The sensitivity s_i of the ith component's reliability R_i is defined as a partial derivation (denoted by f w.r.t. R_i) of system reliability R, denoted as $\Delta_i = \frac{\delta f(R_1, R_2, \ldots R_i, \ldots R_n)}{\delta R_i}$. However, analytical solution is hard when system is large and non-deterministic. In this work, we investigate one component each time. Thus, the formula is then reduced to $\Delta_i = \frac{\delta V(init)}{\delta R_i}$ ($V(init)$ is obtained via reliability distribution). A thread denoting the sensitivity of a particular node can be obtained by solving this equation.

Assumptions and Threads to Validity The reliability analysis mentioned in Figure 4 is based on the probabilistic analysis on Markov models, which shares the same assumptions with the conventional component-based reliability analysis [5, 12, 10, 6]. It assumes that there is statistical independence among failures of the components. Moreover, in our reliability analysis, we assume that any component's failure will eventually

[5] A scheduler in the model denotes for one possible path from the starting node to one accepting node. Since there are non-deterministic choices in the model, there are multiple possible schedulers.

Reliability	UWB	SBTL	SNS	STL	TNO	WiW
Number of Schedulers in MDP	32	24	32	16	64	16
Max Reliability	0.3744	0.4190	0.3670	0.3707	0.3707	0.3707
Min Reliability	0.2956	0.2463	0.2897	0.2927	0.2897	0.2927
Calculation Time	<1 ms					

Table 1: Reliability Prediction

Reliability Requirement	Nodes	UWB	SBTL	SNS	STL	TNO	WiW
0.4	Network	0.854	0.904	0.913	0.911	0.911	0.911
	Sensor	0.886	0.938	0.941	0.923	0.923	0.923
0.5	Network	0.914	-	0.965	0.963	0.963	0.963
	Sensor	0.996	-	0.995	0.994	0.994	0.994
Time(s)		3.45	2.68	3.86	1.87	11.00	2.35

Table 2: Reliability Distribution

result in the failure of the system. Self-recovery/repair scenarios are not considered as failure cases, and these scenarios can be modelled in the MDP [11].

4.2 Reliability Analysis Experiments

Based on the MDP models constructed in Section 3, we use the RaPiD tool for reliability analysis. All the experiments are carried out on a PC with 2.7GHz Intel CPU, 8GB memory and 64-bit Windows 7 operating system. In the following, we listed the settings and results of three groups of experiments respectively. Interested readers are referred to [1] for details.

Reliability Prediction As shown in Table 1, the reliability of six scenarios ranges from 25% to 40% with different scheduler which is quite low considering using the system at home with no human supervision.

One general observation from this experiment is that the system uses the RFID sensors in many places for identity tracking. However, the RFID sensors have the lowest reliability among all the sensors. In fact, due to budget issues, these RFID readers used in the system have a half meter detecting radius which are much cheaper but have a lower accuracy than others with a larger radius. Besides, the dementia patients tend to remove their RFID tags from time to time causing the failure of identity tracking. It is also an important lesson learned that AAL systems cannot rely on patients to provide the critical information. Thus, we suggest the designers to replace the RFID reader to the one with a larger detecting range or the one does not require a tag.

Besides, the six reminding scenarios have similar reliability except for SBTL case. By a careful examination, we discover that the rule defined for SBTL has an error. Because the engineer failed to put the user's identity information into the rule's condi-

tion, this reminder will be sent to the wrong user in 50% chance. This evidenced that reliability analysis is sometimes useful in identifying system bugs.

Reliability Distribution Analysis Further, we explored how to distribute reliability on certain components so as to reach an overall reliability requirement. Two groups of nodes are tested which are sensor nodes and network related nodes. By fixing reliability of the network related nodes, we calculated the distribution on sensor nodes and vice versa. We consider a uniform distribution (where all the nodes have the equal weight) among sensors since they have similar reliability.

As shown in Table 2, it requires each network related node to have a reliability of 0.913 in order for all the scenarios to achieve a reliability of 0.4. However, it is impossible when the requirement raises to 0.5. The reason also points to a failure rule in SBTL scenario. Moreover, it becomes unrealistic that if we expect the system reliability to reach 0.5 based on the current design, it requires highly accurate and stable sensors which are of much higher cost. For example, a short range RFID reader may cost a few hundred US dollars, but for a higher range, the price raises to a few thousand US dollars. Considering the AAL systems are to be deployed in normal homes, the cost becomes unaffordable for normal families. Thus, this group of experiment results requests AMUPADH designers to rethink about the system design rather than simply replace sensors.

Besides, it is still intuitive to ask the question that which node or group of nodes affects the system reliability more than the others? If improvements are made on such node(s), it will be more efficient. Thus, we seek the answer from sensitivity analysis.

Sensitivity Analysis Experiments There are multiple schedulers in each MDP model as shown in Table 2. Due to page limits, we present one typical scheduler in this experiments. The UWB scenario refers to Section 2.2 is modelled in Figure 5. The path connected by thick black links are the target scheduler. It is a typical case which relies on two RFID sensors and multiple other sensors. The iPad case is chosen since playing reminders on iPad is the most common way in practice.

Two nodes and a bundle of nodes are chosen for the experiment which are RFID reader node, Zigbee network node, and bundle of nodes related to Wi-Fi network (If network reliability is improved, the reliability of message transmission paths will also be improved). Figure 6a shows the reliability distribution on these nodes. As we can see, improvement on RFID reader node and Wi-Fi bundle can achieve a higher reliability than Zigbee node. Figure 6b further suggests that when the reliability of these nodes are greater then 0.7, increasing reliability of nodes in Wi-Fi bundle can achieve better improvement than other nodes. In practice, increasing the reliability of a network might be cheaper than purchasing a sensor with higher reliability, e.g., placing more bridges along the path.

4.3 Discussions on Reliability Analysis on AAL Systems

Although we have only carried out detailed analysis on one AAL system, AMUPADH, we contend that the approach considered here is widely applicable to many other similar systems. Our reasons for this assertion are as follows.

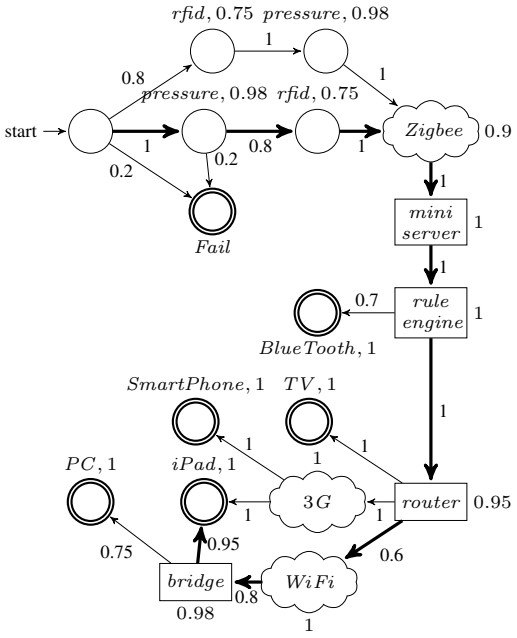
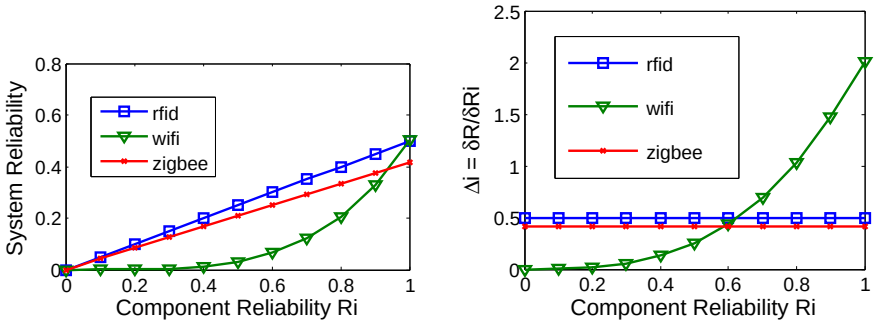


Fig. 5: Bedroom Scenario- UWB: Using Wrong Bed

Usefulness The above experiments show that our approach is able to give a good estimation on the overall system reliability. Upon a reliability requirement on overall system, we are able to provide suggestions of the least requirements on certain nodes. Additionally, our approach provides useful guidance on improving the system effectively, such that relatively more efforts and fund can be spent on those critical components in budget concerned systems. Thus, our approach is able to solve practical problems and give useful suggestions for the improvement of the system design.

Modelling Applicability Layered architecture and multi-sensor platform are widely adopted in AAL systems like AMUPADH system because of the low cost and high extensibility, e.g., plug and play [2]. Thus, the modelling techniques introduced in this case study are easily adaptable to other similar systems by extracting necessary information from the system design and codes about scenarios, transition relations among related sensors and actuators and the internal reasoning mechanism. However, our approach requires the knowledge and experience on modelling system in MDP models, which makes it not easy to use for engineers without necessary background.

Scalability Although some AAL systems may have many more components than AMUPADH system, our approach is scalable. For example, in AMUPADH, there are 9 sensors used in each room while a large AAL system may have more sensors (hundreds of such devices is possible when the AAL system is designed for a building). However, we argue that for a particular scenario to be monitored by the system, only a small portion of the sensors are involved. Otherwise, if many sensors are used, it may not be feasible in reality since the logic of the reasoning sensor input becomes too complicated to



(a) Distribution (b) Sensitivity

Fig. 6: UWB- Sensitivity Analysis on Nodes

be analysed. Furthermore, AAL systems are safety critical that the simpler the system is, the safer it will be. Thus, we conclude that the size of the model and the depth of the probabilistic paths with finite numbers of schedulers are manageable using our approach for current AAL systems. The capability for RaPiD in dealing with large scale models has been studied in [11], which shows it can handle 14K states per second on average in reliability prediction, and reliability distribution and sensitivity analysis is slightly slower due to the workload from the polynomial functions. The improvement of our approach is always possible in future when large case study comes.

5 Conclusion

This paper demonstrate our reliability analysis work of AAL systems AMUPADH using MDP based modelling and verification approach. The models are manually constructed from the design and implementation of the systems. Three groups of experiments are conducted to answer the questions of "What is the overall system reliability with known reliability value of each nodes?", "To reach a certain overall system reliability, how reliable should the sensors/networks be?" and "Which node (could be a sensor or network device) affects the overall reliability the heaviest?". Experiments show surprising results that the overall system reliability is hardly able to reach 50%. It is also suggested that to improve the reliability of Wi-Fi network will be more efficient to improve the system reliability. In future, we will explore methods to automatically generate MDP models. Based on our previous work [15], it is a potential solution to extract an MDP model from the system design model proposed in the paper by labelling the sensor and network devices. Thus, the MDP model can be generated on the fly with the system design model.

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